Name	Type	Feature of Interest	Property	Characteristics	Components	Justification
The name of each IoT device	E.g., sensor, actuator, or device	What "thing" is being observed and/or acted upon?	Which aspect(s) of the "thing" is/are being observed and/or acted upon?	What are the characteristics of the device?	What are the components embedded in the device?	Why is the device selected for the proposed IoT solution?
DFRobot Waterproof DS18B20 Temperature Sensor (BOS0042)	Sensor	Water in each breeding tank	Water temperature	Waterproof DS18B20 probe; 3-5 V; PH2.0-3P connector; sensing range -55 to 125 deg C; display range -10 to 85 deg C; stated error +/-0.5 deg C; pull-up resistor required.	Waterproof DS18B20 probe, 3-pin cable, pull-up resistor, PH2.0 adapter and controller input.	Direct temperature feedback supports heater control and alarm limits in wet tanks. The sealed probe suits remote or humid measurement, while local validation prevents unsafe heating commands.
DFRobot SEN0161-V2 Gravity Analog pH Meter	Sensor	Water in each breeding tank	pH value	Signal-board supply 3.3-5.5 V; analogue output 0-3.0 V; BNC electrode and PH2.0-3P signal connector; stated accuracy +/-0.1 at 25 deg C; electrode range pH 0-14 and 5-60 deg C; response below 2 min; 1 m cable.	Laboratory-grade pH electrode, V2 signal-conversion board, BNC cable, PH2.0-3P lead, calibration buffers and analogue input.	pH is a key water-quality variable for aquaculture. Two-point calibration and local plausibility/rate checks allow the PLC to alarm and bound corrective actions.
DFRobot SEN0189 Turbidity Sensor	Sensor	Water in each breeding tank	Turbidity	Optical infrared emitter/receiver; 5 VDC; 40 mA maximum; response below 500 ms; selectable analogue 0-4.5 V or thresholded digital output; operating range -30 to 80 deg C; probe top is not waterproof.	Optical turbidity probe, interface board, A/D or GPIO input, analogue/digital selector and threshold potentiometer.	Provides early warning of suspended solids, filtration faults and contamination. Digital threshold alarms can be local, while analogue values support trend analysis and water-change decisions.
Lonhand LH-WQS-AN-485-UI NH3-N Sensor	Sensor	Water in each breeding tank	Ammonia-nitrogen concentration	Measurement range 0.1-18,000 mg/L; temperature range 0-60 deg C; 4-20 mA and RS-485 outputs; IP68; 5 m cable; NPT3/4 or M39x1.5 connection; 0-3 bar; pH 4-10; repeatability +/-4%; PC/PBT housing.	Ion-selective ammonia electrode, 4-20 mA/RS-485 transmitter, 5 m cable, immersion fitting, PLC analogue/Modbus input and calibration solution.	Ammonia is a high-risk water-quality condition. A wired industrial output supports continuous trend logging, local alarm thresholds and dependable response when the network is unavailable.
DFRobot Submersible Water Level Sensor (KIT0139)	Sensor	Water in each breeding tank	Water depth and level	0–5 m range; ±0.5% full scale; 4–20 mA; 12–36 VDC; IP68 stainless probe. (DFRobot, n.d.)	Pressure probe, vented cable termination, stilling tube, 24 VDC supply, Opta analogue input	Independent level feedback prevents overflow, excessive drainage and dry-running, and verifies bounded water-change commands.
DFRobot Water Flow Sensor (SEN0217 / YF-S201)	Sensor	Tank inlet and drain pipes	Flow rate and transferred volume	1–30 L/min; Hall-effect pulse output; ±5% over 2–30 L/min; 3.5–12 VDC; 1/2-inch fitting. (DFRobot, n.d.)	Two sensors per tank, service unions, screened low-voltage cable, Opta pulse inputs	Inlet and drain totals cross-check level change and expose blocked pipes, pump faults or stuck valves while limiting exchanged volume.
DFRobot A02YYUW Waterproof Ultrasonic Sensor (SEN0311)	Sensor	Feed in each zone hopper	Remaining feed height	3–450 cm range; 3 cm blind zone; ±1 cm; 100 ms response; IP67; 3.3–5 V UART. (DFRobot, n.d.)	Top mount, dust shield, isolated UART interface, hopper-specific distance-to-fill calibration	One sensor per zone hopper provides a low-feed alert. Calibration accounts for the sloped pellet surface and converts distance to an approximate fill percentage.
Atlas Scientific EZO Humidity Probe (EZO-HUM-P-S)	Sensor	Air in each breeding/nesting zone	Relative humidity and air temperature	0–100% RH; stated ±2% RH from 5–95%; 3.3–5 V; UART/I2C; IP67 stainless probe. (Atlas Scientific, n.d.)	Shielded placement away from spray, isolated interface, mounting bracket, periodic comparison check	Zone humidity reveals condensation risk that can affect electronics and camera visibility, and provides context for nesting-area observations.
Aqua Medic Titanium Heater 500 W	Actuator	Water in each breeding tank	Heat input and water temperature	500 W; 110–240 VAC; intended for about 1000–1500 L; external controller required. (Aqua Medic, n.d.)	Titanium element, rated contactor, RCD, guarded mount, manual isolation, independent over-temperature cut-out	Corrosion-resistant construction suits aquaculture. PLC control uses conservative hysteresis, while the independent cut-out remains the final safety layer.
Bluefish MINI-200 Submersible Pump	Actuator	Tank recirculation and water-change line	Water circulation and flow	12 VDC; 0.4 A; 4.8 W; 200 L/h open flow; 1.5 m maximum head; brushless motor; IP68 submersible construction; approximately 43x38x30 mm; below 35 dB; dry-run/block protection options.	Brushless pump, impeller, inlet/outlet, sealed motor, silicone vibration feet, 12 V supply, relay/MOSFET driver and flow feedback.	Low-voltage circulation supports filtration and controlled water changes. A relay, flow confirmation and dry-run protection prevent a blocked line or empty tank from damaging the pump.
Bürkert Type 6213 EV Normally Closed Valve, 24 VDC	Actuator	Tank inlet and drain lines	Water flow state	Two-way servo-assisted diaphragm valve; normally closed; 24 VDC option; water service; IP65 with suitable plug. (Bürkert, n.d.)	Two valves per tank, selected DN/pressure variant, coil suppression, manual isolation, upstream strainer	Normally closed action fails safe on power loss. Paired with level and flow feedback, it enables bounded partial water exchange.
PondMAX Solar Automatic Fish Feeder	Actuator / device	Feed supplied to turtles in each zone	Feeding time and estimated portion	Solar charging with built-in 2200 mAh lithium battery; 1-6 feedings per day; adjustable S/M/L throw and 0-20 s feed duration; 3-10 mm food size; 12 V/1.66 A adapter specification.	Solar panel, rechargeable battery, hopper, motor and spreader, timer/display controls, protective cover and mounting bracket.	Automated feeding improves schedule consistency and reduces manual labour. Feed commands remain bounded by time, quantity and hopper checks, with operator override and audit records.
AXIS M3086-V Network Camera	Sensor	Turtle activity, feeding and nesting areas	Video and event evidence	4 MP maximum resolution 2688x1512; 2.4 mm F2.1 lens; 130 deg horizontal and 93 deg vertical field of view; H.264/H.265/Motion JPEG; 25/30 fps; WDR up to 120 dB; Lightfinder; PoE Class 2; IP42 and IK08.	Network camera, lens, PoE RJ45 connector, SD-card slot, control button, mounting dome and managed-switch port.	Video provides farm-wide context and event evidence for activity, feeding and nesting. PoE simplifies installation; edge analytics retain only validated clips rather than upload continuous video.